

Lecture 13

PRINCIPLE OF LINEAR IMPULSE AND MOMENTUM AND CONSERVATION OF LINEAR MOMENTUM FOR SYSTEMS OF PARTICLES

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PRINCIPLE OF LINEAR IMPULSE AND MOMENTUM AND CONSERVATION OF LINEAR MOMENTUM FOR SYSTEMS OF PARTICLES

Today's Objectives:

Students will be able to:

1. Calculate the linear momentum of a particle and linear impulse of a force.
2. Apply the principle of linear impulse and momentum.



In-Class Activities:

- Reading Quiz
- Applications
- Principle of Linear Impulse and Momentum
- Conservation of Linear Momentum for a System of Particles
- Example Problems
- Concept Quiz
- Group Problem Solving
- Attention Quiz

READING QUIZ

1. The linear impulse and momentum equation is obtained by integrating the _____ with respect to time.

A) friction force B) equation of motion
C) kinetic energy D) potential energy
2. Which parameter is not involved in the linear impulse and momentum equation?

A) Velocity B) Displacement
C) Time D) Force

APPLICATIONS



A dent in a trailer fender can be removed using an impulse tool, which delivers a force over a very short time interval. To do so, the weight is gripped and jerked upwards, striking the stop ring.

How can we determine the magnitude of the linear impulse applied to the fender?

Could you analyze a carpenter's hammer striking a nail in the same fashion?

Sure!

APPLICATIONS (continued)



When a stake is struck by a sledgehammer, a large impulse force is delivered to the stake and drives it into the ground.

If we know the initial speed of the sledgehammer and the duration of impact, how can we determine the magnitude of the impulsive force delivered to the stake?

PRINCIPLE OF LINEAR IMPULSE AND MOMENTUM

(Section 15.1)

The next method we will consider for solving particle kinetics problems is obtained by **integrating the equation of motion with respect to time**.

The result is referred to as the **principle of impulse and momentum**. It can be applied to problems involving both linear and angular motion.

This principle is useful for solving problems that involve **force, velocity, and time**. It can also be used to analyze the mechanics of **impact**.

PRINCIPLE OF LINEAR IMPULSE AND MOMENTUM (continued)

The principle of linear impulse and momentum is obtained by integrating the equation of motion with respect to time. The equation of motion can be written

$$\sum \mathbf{F} = m \mathbf{a} = m (d\mathbf{v}/dt)$$

Separating variables and integrating between the limits $\mathbf{v} = \mathbf{v}_1$ at $t = t_1$ and $\mathbf{v} = \mathbf{v}_2$ at $t = t_2$ results in

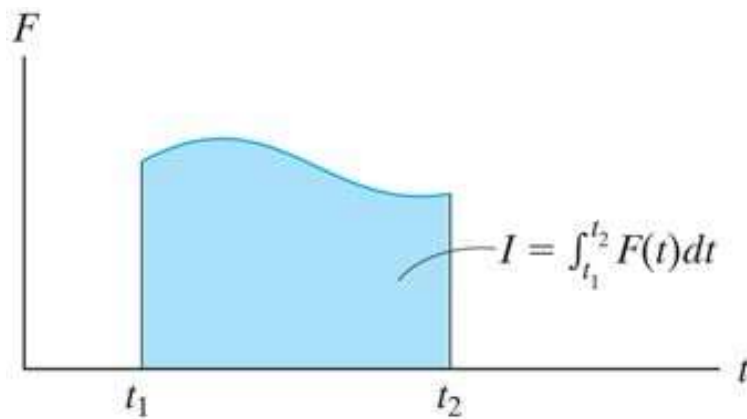
$$\sum \int_{t_1}^{t_2} \mathbf{F} dt = m \int_{\mathbf{v}_1}^{\mathbf{v}_2} d\mathbf{v} = m\mathbf{v}_2 - m\mathbf{v}_1$$

This equation represents the principle of linear impulse and momentum. It relates the particle's final velocity ($\mathbf{v}_2$) and initial velocity ($\mathbf{v}_1$) and the forces acting on the particle as a function of time.

PRINCIPLE OF LINEAR IMPULSE AND MOMENTUM (continued)

Linear momentum: The vector $m\mathbf{v}$ is called the linear momentum, denoted as $\mathbf{L}$. It is a **vector quantity** measuring the mass in motion. $\mathbf{L}$ acts in the **same direction** as $\mathbf{v}$ and has units of $(\text{kg}\cdot\text{m})/\text{s}$.

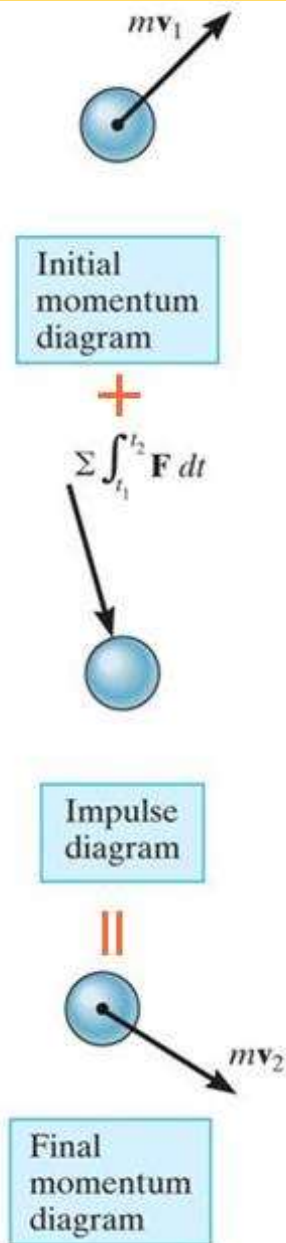
Linear impulse: The integral $\int \mathbf{F} dt$ is the linear impulse, denoted $\mathbf{I}$. It is a **vector quantity** measuring the effect of a force during its time interval of action. $\mathbf{I}$ acts in the **same direction** as $\mathbf{F}$ and has units of $\text{N}\cdot\text{s}$.



The impulse may be determined by **direct integration**. Graphically, it can be represented by the **area under the force versus time curve**. If $\mathbf{F}$ is constant, then

$$\mathbf{I} = \mathbf{F} (t_2 - t_1) .$$

PRINCIPLE OF LINEAR IMPULSE AND MOMENTUM (continued)



The principle of linear impulse and momentum in **vector** form is written as

$$m\mathbf{v}_1 + \sum \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2$$

The particle's initial momentum plus the sum of all the impulses applied from t_1 to t_2 is equal to the particle's final momentum.

The two **momentum diagrams** indicate direction and magnitude of the particle's initial and final momentum, $m\mathbf{v}_1$ and $m\mathbf{v}_2$. The **impulse diagram** is similar to a free body diagram but includes the time duration of the forces acting on the particle.

IMPULSE AND MOMENTUM: SCALAR EQUATIONS

Since the principle of linear impulse and momentum is a vector equation, it can be resolved into its x, y, z component scalar equations:

$$m(v_x)_1 + \sum \int_{t_1}^{t_2} F_x dt = m(v_x)_2$$

$$m(v_y)_1 + \sum \int_{t_1}^{t_2} F_y dt = m(v_y)_2$$

$$m(v_z)_1 + \sum \int_{t_1}^{t_2} F_z dt = m(v_z)_2$$

The scalar equations provide a convenient means for applying the principle of linear impulse and momentum once the velocity and force vectors have been resolved into x, y, z components.

PROBLEM SOLVING

- Establish the x, y, z coordinate system.
- Draw the particle's free body diagram and establish the direction of the particle's initial and final velocities, drawing the impulse and momentum diagrams for the particle. Show the linear momenta and force impulse vectors.
- Resolve the force and velocity (or impulse and momentum) vectors into their x, y, z components, and apply the principle of linear impulse and momentum using its scalar form.
- Forces as functions of time must be integrated to obtain impulses. If a force is constant, its impulse is the product of the force's magnitude and time interval over which it acts.

CONSERVATION OF LINEAR MOMENTUM FOR A SYSTEM OF PARTICLES (Section 15.3)



When the **sum of external impulses** acting on a system of objects is **zero**, the linear impulse-momentum equation simplifies to

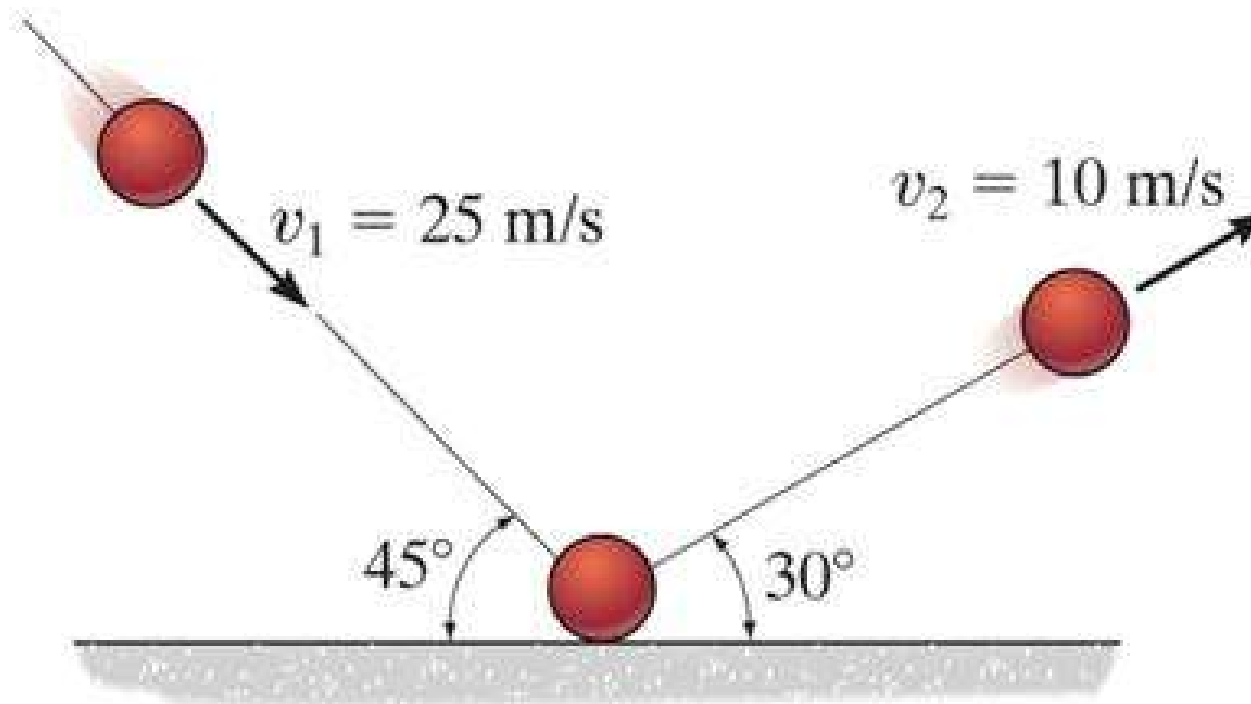
$$\sum m_i(\mathbf{v}_i)_1 = \sum m_i(\mathbf{v}_i)_2$$

This equation is referred to as the **conservation of linear momentum**. **Conservation of linear momentum is often applied when particles collide or interact**. When particles impact, only impulsive forces cause a change of linear momentum.

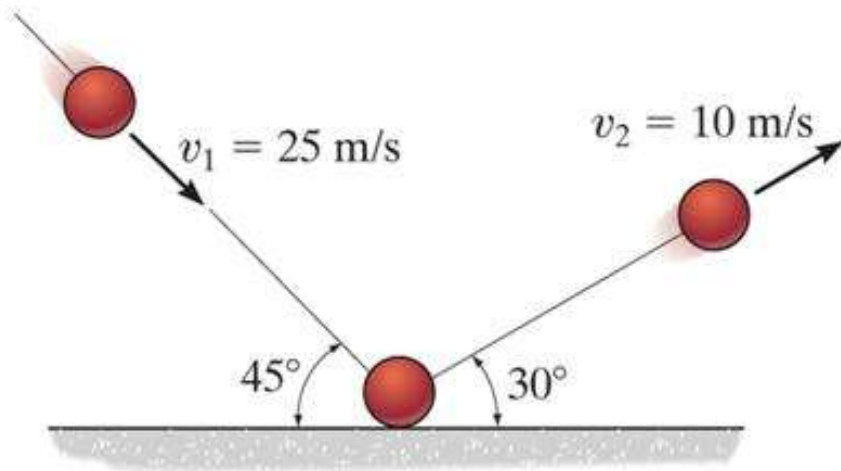
The sledgehammer applies an impulsive force to the stake. The weight of the stake is considered negligible, or non-impulsive, as compared to the force of the sledgehammer. Also, provided the stake is driven into soft ground with little resistance, the impulse of the ground acting on the stake is considered non-impulsive.

EXAMPLE I

The 0.5-kg ball strikes the rough ground and rebounds with the velocities shown. Determine the magnitude of the impulse the ground exerts on the ball. Assume that the ball does not slip when it strikes the ground, and neglect the size of the ball and the impulse produced by the weight of the ball. **(Problem No. F15-1)**



EXAMPLE I



Given: A 0.5 kg ball strikes the rough ground and rebounds with the velocities shown. Neglect the ball's weight during the time it impacts the ground.

Find: The magnitude of impulsive force exerted on the ball.

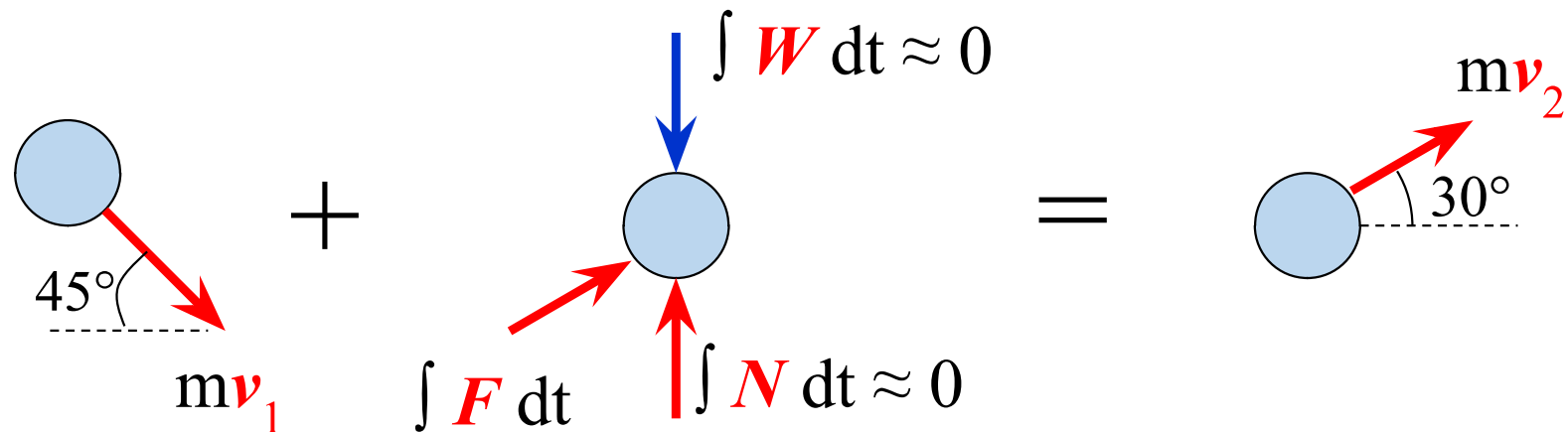
Plan:

- 1) Draw **momentum and impulse diagrams** of the ball as it hits the surface.
- 2) Apply the principle of impulse and momentum to determine the impulsive force.

EXAMPLE I (continued)

Solution:

1) Impulse and momentum diagrams can be drawn as:



The impulse caused by the ball's weight and the normal force $\mathbf{N}$ can be neglected because their magnitudes are very small as compared to the impulse from the ground.

EXAMPLE I (continued)

- 2) The principle of impulse and momentum can be applied along the direction of motion:

$$m\mathbf{v}_1 + \sum \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2$$

$$\Rightarrow 0.5 (25 \cos 45^\circ \mathbf{i} - 25 \sin 45^\circ \mathbf{j}) + \int_{t_1}^{t_2} \sum \mathbf{F} dt = 0.5 (10 \cos 30^\circ \mathbf{i} + 10 \sin 30^\circ \mathbf{j})$$

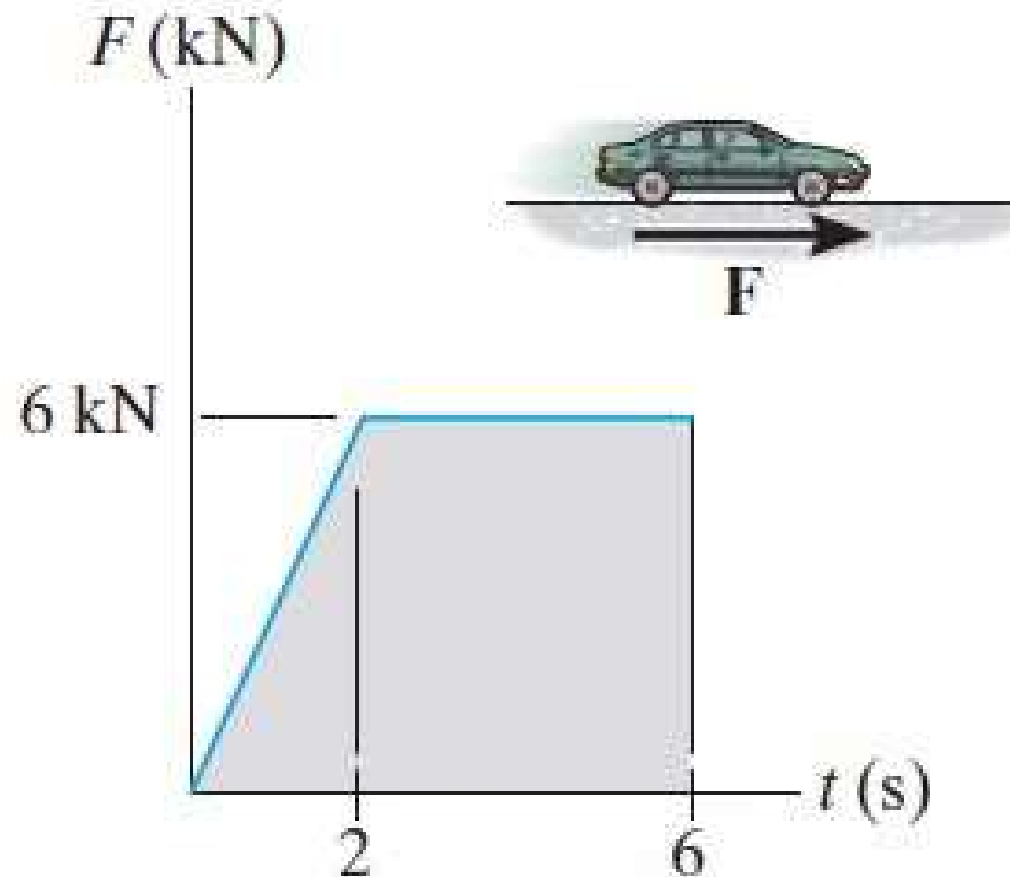
The impulsive force vector is

$$\mathbf{I} = \int_{t_1}^{t_2} \sum \mathbf{F} dt = (4.509 \mathbf{i} + 11.34 \mathbf{j}) \text{ N} \cdot \text{s}$$

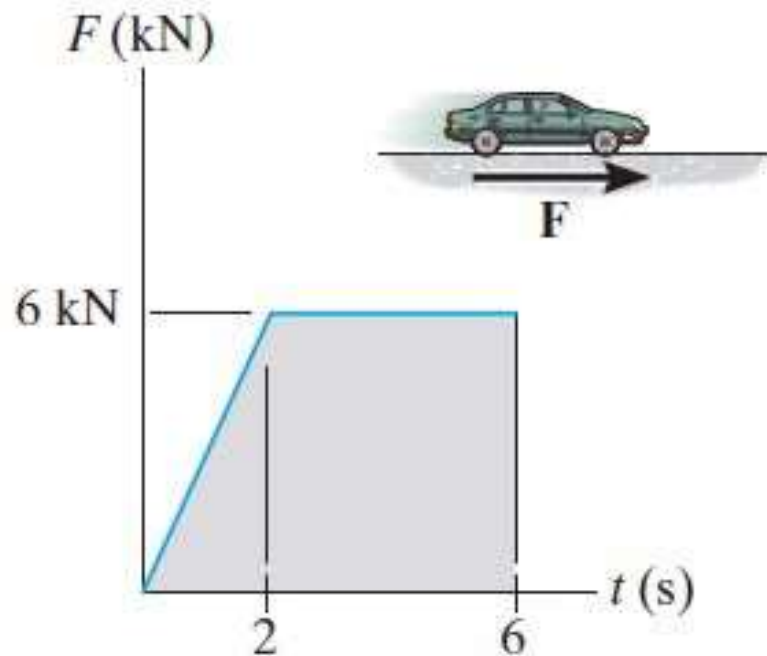
$$\text{Magnitude: } I = \sqrt{4.509^2 + 11.34^2} = \underline{12.2 \text{ N} \cdot \text{s}}$$

EXAMPLE II

The wheels of the 1.5-Mg car generate the traction force F described by the graph. If the car starts from rest, determine its speed when $t = 6$ s. **(Problem No. F15-4)**



EXAMPLE II



Given: The wheels of the 1500 kg car generate a traction force F described by the graph.

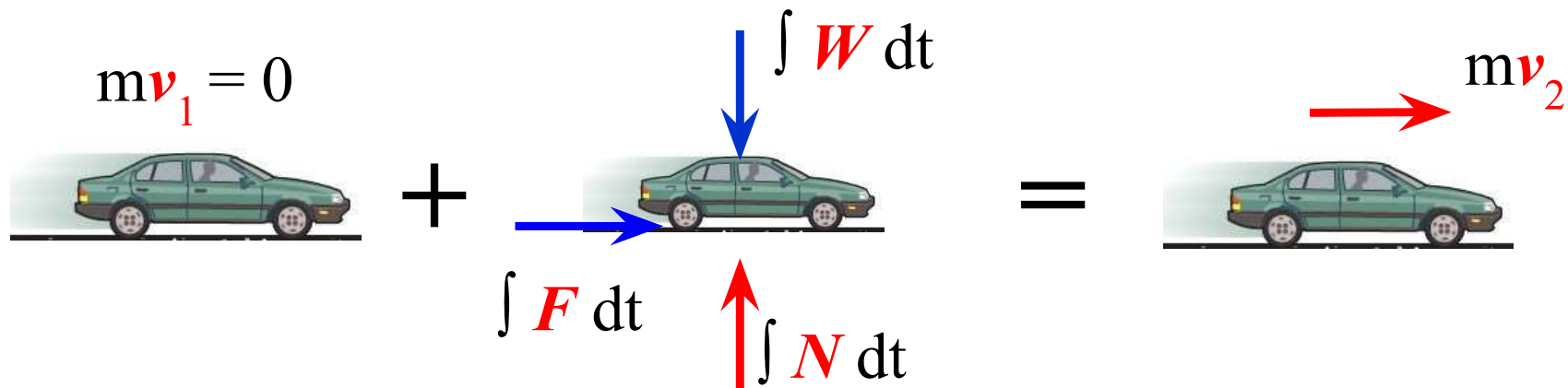
Find: The speed of the car when $t = 6$ s if the car starts from rest.

- Plan:**
- 1) Draw the **momentum and impulse diagrams** of the car.
 - 2) Apply the principle of impulse and momentum to determine the speed.

EXAMPLE II (continued)

Solution:

1) The impulse and momentum diagrams can be drawn as:



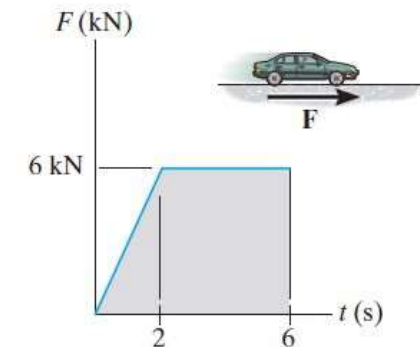
Why is the traction force $\mathbf{F}$ drawn acting to the right?

The impulse caused by the weight $\mathbf{W}$ and the normal force $\mathbf{N}$ can be canceled. Why?

EXAMPLE II (continued)

2) The principle of impulse and momentum can be applied along the direction of motion:

$$+ \rightarrow m\mathbf{v}_1 + \sum \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2$$



$$(1500 \text{ kg}) (0 \text{ m/s}) + 0.5 (6000 \text{ N}) (2 \text{ s}) + (6000 \text{ N}) (4 \text{ s}) \\ = (1500 \text{ kg}) (v_2 \text{ m/s})$$

$$v_2 = \underline{20 \text{ m/s}}$$

CONCEPT QUIZ

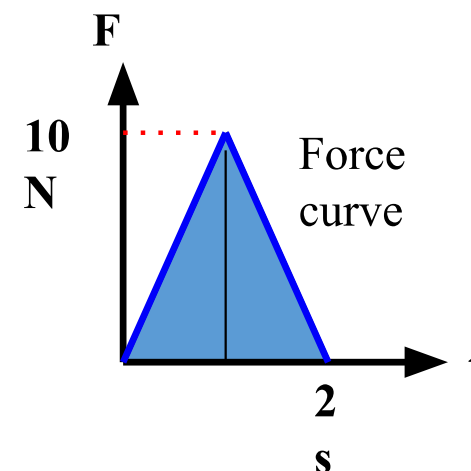
1. Calculate the impulse due to the force.

A) $20 \text{ kg}\cdot\text{m/s}$

B) $10 \text{ kg}\cdot\text{m/s}$

C) $5 \text{ N}\cdot\text{s}$

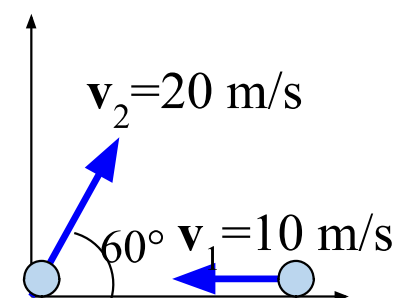
D) $15 \text{ N}\cdot\text{s}$



2. A constant force $\mathbf{F}$ is applied for 2 s to change the particle's velocity from $\mathbf{v}_1$ to $\mathbf{v}_2$. Determine the force $\mathbf{F}$ if the particle's mass is 2 kg.

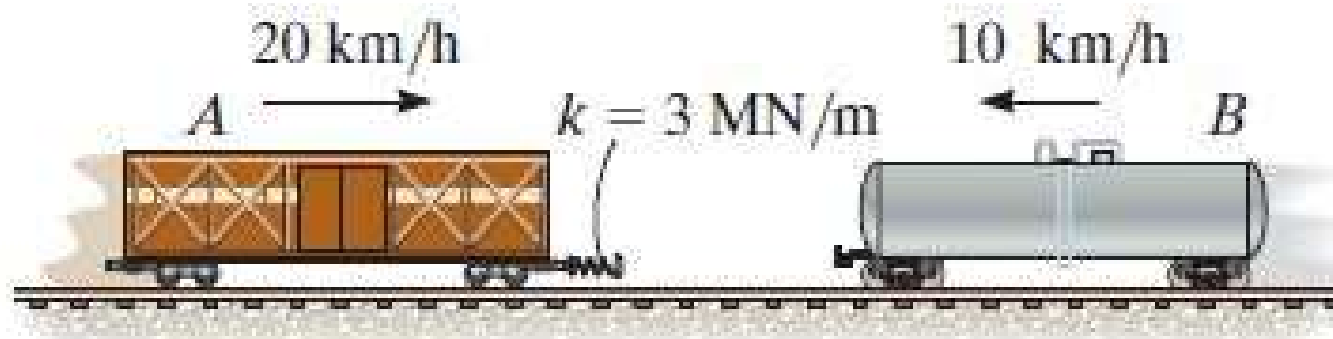
A) $(17.3 \mathbf{j}) \text{ N}$ B) $(-10 \mathbf{i} + 17.3 \mathbf{j}) \text{ N}$

C) $(20 \mathbf{i} + 17.3 \mathbf{j}) \text{ N}$ D) $(10 \mathbf{i} + 17.3 \mathbf{j}) \text{ N}$



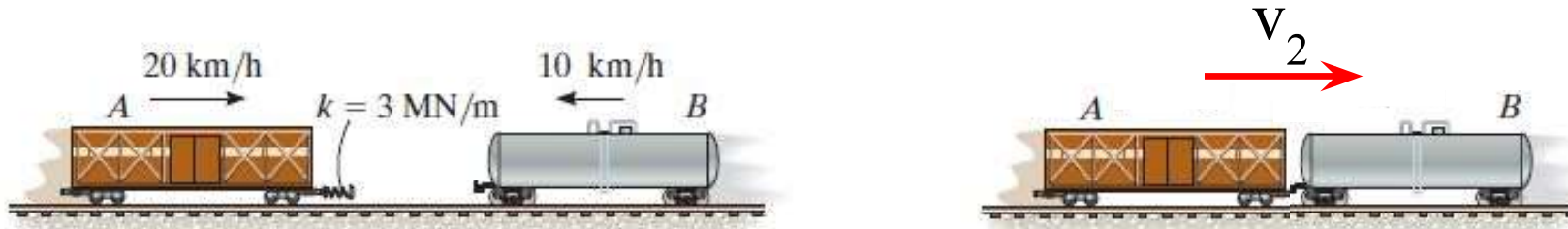
GROUP PROBLEM SOLVING

The 30-Mg freight car A and 15-Mg freight car B are moving towards each other with the velocities shown. Determine the maximum compression of the spring mounted on car A. Neglect rolling resistance. **(Problem No. 15-47)**



GROUP PROBLEM SOLVING (continued)

Solution:



Conservation of linear momentum along x-axis:

$$m_A(v_{A1}) + m_B(v_{B1}) = (m_A + m_B) v_2$$

$$\text{where } v_{A1} = 20 \text{ km/h} = 5.556 \text{ m/s,}$$

$$v_{B1} = -10 \text{ km/h} = -2.778 \text{ m/s}$$

$$30,000 (5.556) + 15,000 (-2.778) = (30,000 + 15,000) v_2$$

$$v_2 = \underline{2.778 \text{ m/s}} \rightarrow$$

GROUP PROBLEM SOLVING (continued)

Then, **energy conservation equation** to find the maximum compression of the spring ;

$$\begin{aligned} T_1 + V_1 &= T_2 + V_2 \\ [0.5 (30000) (5.556)^2 + 0.5 (15000) (-2.778)^2] + 0 \\ &= 0.5 (45000) (2.778)^2 + 0.5 (3 \times 10^6) (s_{\max})^2 \end{aligned}$$

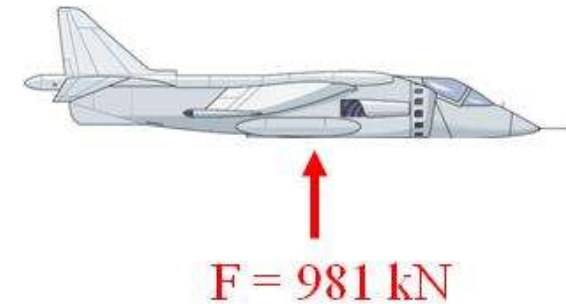
The maximum compression of the spring:

$$s_{\max} = 0.4811 \text{ m} = \underline{481 \text{ mm}}$$

ATTENTION QUIZ

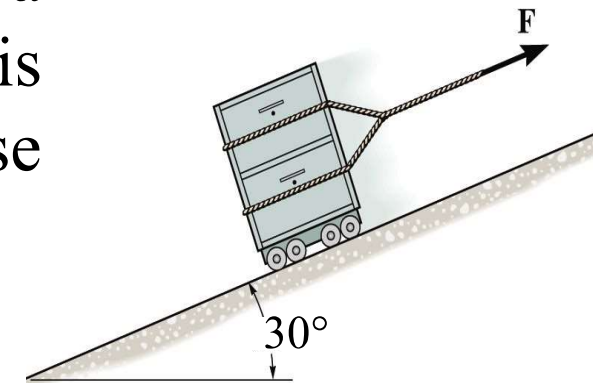
1. Jet engines on the 100 Mg VTOL aircraft exert a constant vertical force of 981 kN as it hovers. Determine the net impulse on the aircraft over $t = 10$ s.

- A) $-981 \text{ kN}\cdot\text{s}$ B) $0 \text{ kN}\cdot\text{s}$
C) $981 \text{ kN}\cdot\text{s}$ D) $9810 \text{ kN}\cdot\text{s}$



2. A 100-N (≈ 10 -kg) cabinet is placed on a smooth surface. If a force of a 100 N is applied for 2 s, determine the net impulse on the cabinet during this time interval.

- A) $0 \text{ N}\cdot\text{s}$  B) $100 \text{ N}\cdot\text{s}$ 
C) $200 \text{ N}\cdot\text{s}$  D) $300 \text{ N}\cdot\text{s}$ 



End of the Lecture
Let The Learning Continue